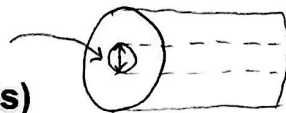


COURSE: CHE 312: Transport Phenomena II**Fall Semester 2025****Department of Chemical Engineering****University of Illinois Chicago****Quiz - Exam****Duration: 70 minutes --- October 23, 2025****Both Problems must be done**

$$D = 0.5 \text{ in}$$



$$A = 2\pi RL$$

$$r = \frac{D}{2} = 0.25 \text{ in} \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 0.0208 \text{ ft}$$

Problem 1: (50 points)

Let's design a shell and tube heat exchanger to heat a stream with $m_C C_{p,C} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_H C_{p,H} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

$$q = \dot{m}_H C_{p,H} (T_{H,i} - T_{H,o}) = \dot{m}_C C_{p,C} (T_{C,o} - T_{C,i}) \Rightarrow \left(5,000 \frac{\text{BTU}}{\text{h}^\circ\text{F}}\right) (210^\circ\text{F} - 90^\circ\text{F}) = \left(10,000 \frac{\text{BTU}}{\text{h}^\circ\text{F}}\right) (260^\circ\text{F} - T_{H,o})$$

$$q = U_o A \Delta T_m = U_o A \left(\frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)} \right)$$

$$\rightarrow 600,000 \frac{\text{BTU}}{\text{h}} = 2,600,000 \frac{\text{BTU}}{\text{h}} - 10,000 \frac{\text{BTU}}{\text{h}^\circ\text{F}} T_{H,o}$$

$$\Rightarrow 10,000 \frac{\text{BTU}}{\text{h}^\circ\text{F}} (T_{H,o}) = 2,000,000 \frac{\text{BTU}}{\text{h}} \Rightarrow T_{H,o} = 200^\circ\text{F}$$

$$q = U_o (2\pi RL) \left(\frac{\Delta T_2 - \Delta T_1}{\ln(\Delta T_2 / \Delta T_1)} \right) \cdot N_t; q = \left(5,000 \frac{\text{BTU}}{\text{h}^\circ\text{F}}\right) (210^\circ\text{F} - 90^\circ\text{F}) = 600,000 \frac{\text{BTU}}{\text{h}}$$

$$L = \frac{q \ln(\Delta T_2 / \Delta T_1)}{U_o (2\pi R) (\Delta T_2 - \Delta T_1) \cdot N_t} = \frac{\left(600,000 \frac{\text{BTU}}{\text{h}}\right) \ln(60/120)}{\left(150 \frac{\text{BTU}}{\text{h ft}^2 \text{ }^\circ\text{F}}\right) (2\pi (0.0208 \text{ ft})) (-60^\circ\text{F}) (30)} \Rightarrow L = 11.78 \text{ ft}$$

b.) If the tubes were shorter, in order to compensate for this, the number of tubes would need to be increased

$$L = \frac{q(\ln(\Delta T_2 - \Delta T_1))}{U_o(2\pi R)(\Delta T_2 - \Delta T_1) \cdot N_t}$$

Since this
is smaller,

denominator must
be larger to compensate

$$m_{\text{tot}} = 1 \text{ kg} \quad R = 0.01 \text{ m} \quad T_{i,s} = 90^\circ\text{C} \quad , \quad T_{f,s} = 40^\circ\text{C} \quad , \quad T_w = 20^\circ\text{C}$$

$$V_c = 1 \text{ L}$$

Problem 2: (60 points)

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10 \text{ mm}$ initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500 \text{ W/m}^2\text{K}$. $B_i \ll 0.1$
The density of aluminum is $\rho = 2,700 \text{ kg/m}^3$ and its heat capacity is $C_p = 900 \text{ J/kg K}$. Assume that the water temperature remains constant. (30 points)

a.) If the changing temperature of the water is considered, determine the time needed to cool the aluminum spheres to 40°C . (30 points) $A = 4\pi R^2$

Assuming Lumped Capacitance for a single, 1 kg sphere of Aluminum:

$$-h A_s (T - T_w) = \rho V_c \frac{dT}{dt} \Rightarrow \int_0^t \frac{-h A_s}{\rho V_c} dt = \int_{T_{i,s}}^{T_{f,s}} (T - T_w)^{-1} dT \Rightarrow -\frac{h A_s}{\rho V_c} t = \ln\left(\frac{T_{f,s} - T_w}{T_{i,s} - T_w}\right)$$

$$\Rightarrow t = -\frac{\rho V_c}{h A_s} \ln\left(\frac{T_{f,s} - T_w}{T_{i,s} - T_w}\right) = -\frac{(2700 \frac{\text{kg}}{\text{m}^3})(0.001 \text{ m}^3)}{(500 \frac{\text{W}}{\text{m}^2\text{K}})(4\pi(0.01 \text{ m})^2)} \ln\left(\frac{40^\circ\text{C} - 20^\circ\text{C}}{90^\circ\text{C} - 20^\circ\text{C}}\right) = 5.38 \frac{\text{kg}\cdot\text{K}\cdot\text{s}}{\text{J}}$$

$$5.38 \frac{\text{kg}\cdot\text{K}\cdot\text{s}}{\text{J}} \cdot \underset{(C_p)}{900 \frac{\text{J}}{\text{kg}\cdot\text{K}}} = 4845 \text{ s} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \boxed{80.75 \text{ minutes}}$$

b.) $\frac{d\theta^*}{dt^*} = \frac{1}{(r^*)^2} \frac{d}{dr} \left((r^*)^2 \frac{d\theta^*}{dr^*} \right)$ where $r^* = \frac{r}{r_0}$, $t^* = \frac{\alpha t}{r_0^2}$
 $\alpha = \frac{k}{\rho C_p}$, $\theta^* = \frac{T - T_i}{T_s - T_\infty}$
 why make the problem more difficult?

This looks like fun to solve, but I'd need a day or two :)

$$m_{c,i} c_{p,c} \Delta T_m = m_{h,i} c_{p,h} \Delta T_m$$

$$UA \Delta T_m$$